

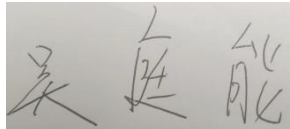
保密等级：机密

SPECIFICATION

产品规格书

SKI.WB800D80S.1 C23365

IEEE 802.11 a/b/g/n/ac/ax 1T1R USB+SDIO Wi-Fi Integrated Bluetooth 2.1+EDR/3.0/4.x/5.2/5.4

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REVISION HISTORY

VERSION	DATE	BOARD ID	PAGE	DESCRIPTION	AUTHOR
V1.0	2023.02.27	SKI.WB800D80S.1 B23082	ALL	First Issued	Lvguitian
V1.1	2023.03.10	SKI.WB800D80S.1 B23082	ALL	更新射频参数	Lvguitian
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V1.7	2023.12.26	SKI.WB800D80S.1 C23365	24	更新包材信息	Lvguitian
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V2.0	2024.8.16	SKI.WB800D80S.1 C23365	5~6	增加料号	Lvguitian
V2.1	2024.12.10	SKI.WB800D80S.1 C23365	9	修正工作温度范 围	Lvguitian

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1. Introduction (简介)

SKI.WB800D80S.1 module is based on AIC8800D80 solution. SKI.WB800D80S.1 is a WiFi6/BT5.4 combo low-power, high-performance and high-integrated dual band wireless communication module which is designed for meeting the customers' needs of small size and low cost. This module supports both WLAN and BT functions. Its WLAN/BT function supports the USB2.0/SDIO3.0 interface, and its BT function supports the UART interface, and the module meets the requirements of standard protocol IEEE 802.11 a/b/g/n/ac/ax. Such units as power management, power amplifier and low-noise amplifier are integrated in the main chip of the module. Its WLAN PHY rate is up to 600.4Mbps@TX. The module can be applied in smart sound boxes, set-top boxes, game machines, printers, IP cameras, tachographs, and other smart equipment. This documentation describes the engineering requirements specification.

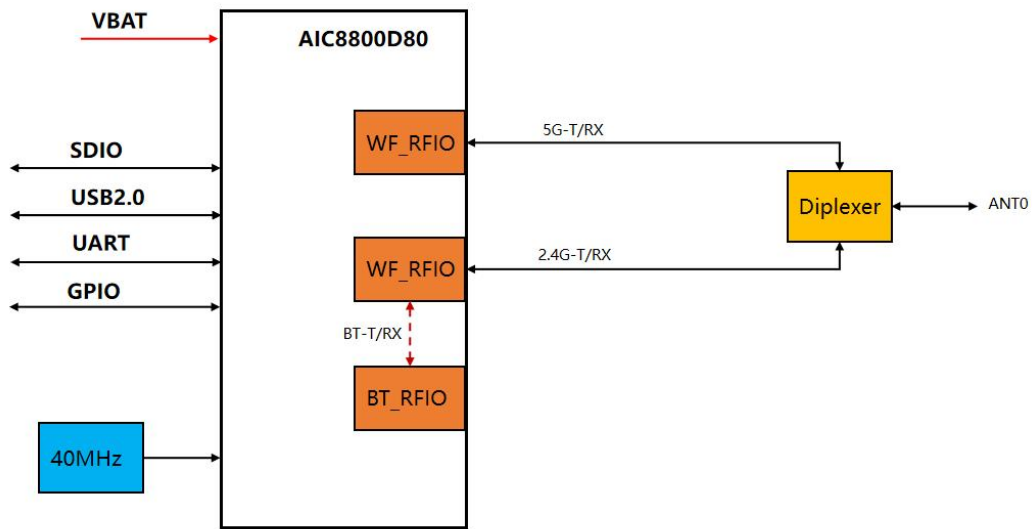
SKI.WB800D80S.1 模块基于 AIC8800D80 方案。SKI.WB800D80S.1 是一款 Wi-Fi6/BT5.4

组合的低功耗、高性能、高集成度双频无线通信模块，专为满足客户小尺寸、低成本的需求而设计。该模块支持WLAN 和BT 功能。WLAN/BT 功能支持USB2.0 / SDIO3.0 接口，BT 功能支持UART接口，满足 IEEE 802.11 a/b/g/n/ac/ax 标准协议要求。本文档描述了工程要求规范。

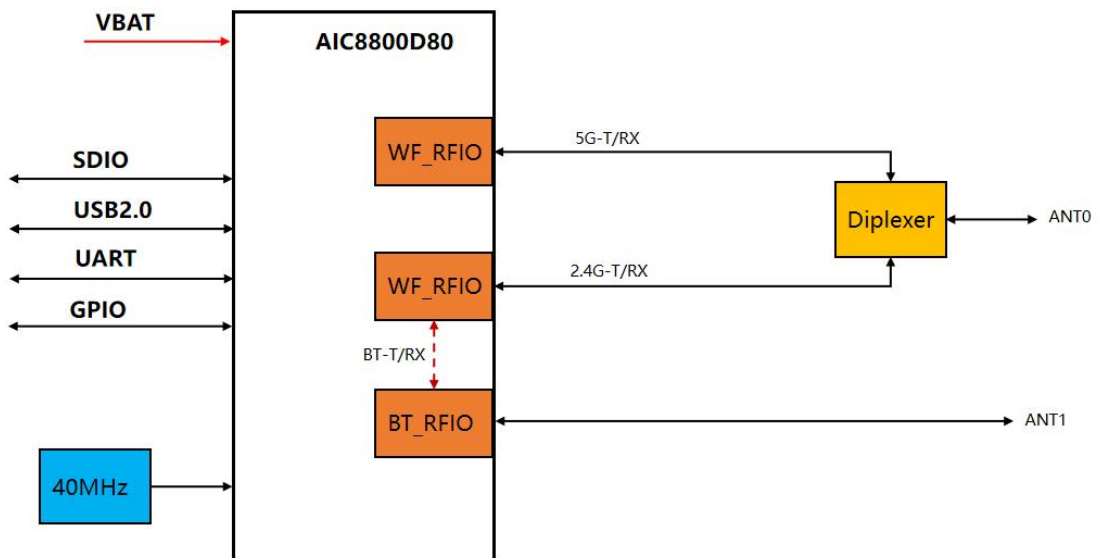
2. Features (特性)

Reserving System 接收制式	IEEE Std. 802.11a
	IEEE Std. 802.11b
	IEEE Std. 802.11g
	IEEE Std. 802.11n
	IEEE Std. 802.11ac
	IEEE Std. 802.11ax
	Bluetooth 2.1+EDR/3.0/4.x/5.2/5.4
Chip Solution 芯片方案	AIC8800D80
Band 波段	2.4GHz/5G
Bandwidth 带宽	20/40/80MHz
Dimensions 尺寸	12mm×12mm×1.9mm
Antenna 天线	Stamp Hole
Installation Mode 安装方式	SMD
Remark 备注	

3. Block Diagram (结构框图)

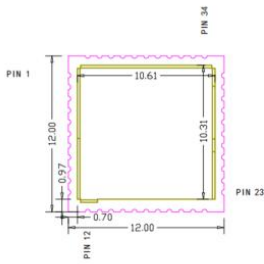


SKI.WB800D80S.1单天线配置系统框图

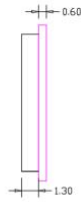


SKI.WB800D80S.1双天线配置系统框图

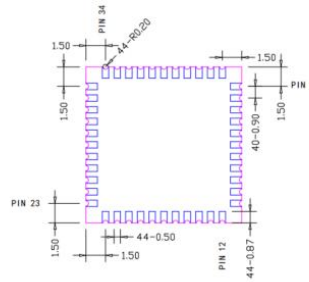
4. Package Outline and Mounting (外形及安装尺寸)



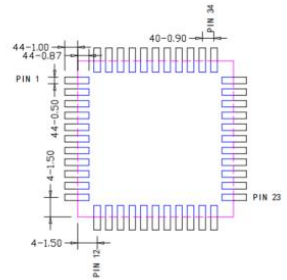
模组主视图



模组侧视图



模组后视图



建议上板封装尺寸

NOTE:

1. 板内顶层器件最高1.5mm，底层无器件；
2. 模组外形尺寸公差为 $\pm 0.15\text{mm}$ ，板厚以及未标注公差为 $\pm 0.1\text{mm}$ 。

5. Pin Definition（引脚定义）



PIN	SYMBOL	DESCRIPTION
1	GND	Connected to Ground
2	ANT0	2.4&5G&BT 天线接口(单天线配置) 2.4&5G 天线接口(双天线配置)
3	GND	Connected to Ground
4	ANT1	NC(单天线配置) BT 天线接口(双天线配置)
5	NC	-
6	GPIOB3/HOST_WAKE_BT	通用GPIO/主机唤醒蓝牙
7	GPIOB2/BT_WAKE_HOST	通用GPIO/蓝牙唤醒主机
8	NC	-
9	VBAT	3.3V 供电
10	USB_DM	USB 功能
11	USB_DP	USB 功能
12	PWR_WF	使能引脚，内部上拉47Kohm
13	GPIOB1/WL_WAKE_HOST	通用GPIO/WIFI唤醒主机
14	SDIO_D2	I/O
15	SDIO_D3	I/O
16	SDIO_CMD	I/O
17	SDIO_CLK	I/O
18	SDIO_D0	I/O
19	SDIO_D1	I/O
20	GND	Connected to Ground
21	NC	-
22	VIO	3.3V/1.8V，建议选用1.8V
23	NC	-
24	GPIOB0/HOT_WAKE_WL	通用GPIO/主机唤醒WIFI
25	PCM_OUT	I/O
26	PCM_CLK	I/O
27	PCM_IN	I/O

28	PCM_SYNC	I/O
29	UART0_TX	I/O
30	UART0_RX	I/O
31	GND	Connected to Ground
32	NC	-
33	GND	Connected to Ground
34	NC	NC
35	NC	NC
36	GND	Connected to Ground
37	NC	-
38	NC	-
39	NC	-
40	NC	-
41	UART1_RTS	UART1_RTS
42	UART1_TX	UART1_TX
43	UART1_RX	UART1_RX
44	UART1_CTS	UART1_CTS

6. Product Pictures（实物图片）

模组料号	正视图	背视图	备注
002.029.0001170			双天线/带屏蔽罩
002.029.0001214			双天线/带屏蔽罩 客户定制标签
002.029.0001231			双天线/带屏蔽罩 客户定制标签
002.029.0001271			双天线/不带屏蔽罩
002.029.0001351			单天线/带屏蔽罩

丝印说明:

- (1) 红色方框内的字符为产品 PCB 型号;
- (2) 黄色方框内的字符为产品周期号;
- (3) 其他为非关键字符, 无须管控。

7. Key Materials (关键物料)

序号	关键件名称	型号	规格/材料	备注
1	集成电路	AIC8800D80	QFN48	AIC
2	PCB	SKI.WB800D80S.1	FR-4,4LAY	
3	晶体振荡器	40MHz, $\pm 10\text{ppm}$, 2016	40MHz	JWT,MDH,TXC,Murata
4	双工器	SLFD18-5R950G-07T FLT18D254959D-3206A	2.4GHz/5GHz	Sunlord FTR

8. General Requirements (一般要求)

No.	Feature	Description
8-1	Operation Voltage 工作电压范围	$3.3\text{V} \pm 0.3$
8-2	Current Consumption 最大电流	600mA
8-3	Ripple 纹波	$\leq 120\text{mV}$
8-4	Operation Temperature 工作温度范围	-20°C to $+70^{\circ}\text{C}$
8-5	Antenna Type 天线类型	External antenna
8-6	Interface	SDIO3.0/USB2.0/PCM/UART
8-7	Storage Temperature 存储温度	-40°C to $+85^{\circ}\text{C}$

*纹波：定义是主板给模组供电处的纹波要求

*RFPower Configuration Requirements (射频功率配置需求)

txpwr_index_2.4g: [0]:11b+11a/g, [1]:11n/11ac, [2]:11ax

[0] = 20 20 20 20 18 18 18 18 16 16 15 15

[1] = 18 18 18 18 16 16 15 15 14 14

[2] = 18 18 18 18 16 16 15 15 14 14 13 13

txpwr_index_5g: [0]:11a, [1]:11n/11ac, [2]:11ax

[0] = 18 18 18 18 16 16 15 15

[1] = 18 18 18 18 16 16 15 15 14 14

[2] = 18 18 18 18 16 16 14 14 13 13 12 12

*上板需按照上述的要求配置 **userconfig** 值表,这样才是模组的实际的功率值,若改变上面的值一个值相对提高或者降低 **1dBm**, 调控要求:根据上板测试性能要求,可相对调控。

9. Electrical Characteristics (电气特性)

除非另有说明，电气规范试验都在下列条件下进行：

环境条件温度：25℃±5℃；

电源电压：模块输入电压 3.3V（±10%）；

The Test for electrical specification was performed under the following condition unless otherwise specified.

Ambient condition Temperature :25℃ ± 5℃;

Power supply voltages: 3.3V (±10%) input power at the Module;

9.1 IEEE 802.11b Section (2.4GHz)

Items	Contents				
Specification	IEEE802.11b				
Mode	CCK				
Channel	CH1 to CH13				
Data rate	1, 2, 5.5, 11Mbps				
TX Characteristics	Min.	Typ.	Max.	Unit	Remark
1. Power Levels at each rate	18	20	22	dBm	
(1Mbps~11Mbps)					
2. Spectrum Mask @ target power					
1) fc ±11MHz to ±22MHz	-	-	-30	dBr	
2) fc > ±22MHz	-	-	-50	dBr	
3. Constellation Error(EVM)@ target power					
1) 1Mbps	-	-	-9.11	dB	
2) 2Mbps	-	-	-9.11	dB	
3) 5.5Mbps	-	-	-9.11	dB	
4) 11Mbps	-	-	-9.11	dB	
4. Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
5. Minimum Input Level Sensitivity					
1) 1Mbps (FER ≤8%)	-	-	-83	dBm	
2) 2Mbps (FER ≤8%)	-	-	-80	dBm	
3) 5.5Mbps (FER ≤8%)	-	-	-79	dBm	
4) 11Mbps (FER ≤8%)	-	-	-76	dBm	
6. Maximum Input Level (FER ≤8%)	-10	-	-	dBm	

9.2 IEEE 802.11g Section (2.4GHz)

Items	Contents				
Specification	IEEE802.11g				
Mode	OFDM				
Channel	CH1 to CH13				
Data rate	6, 9, 12, 18, 24, 36, 48, 54Mbps				
TX Characteristics	Min.	Typ.	Max.	Unit	Remark
1. Power Levels					
1) For antenna port (6~18M)	16	18	20	dBm	
2) For antenna port (24~36M)	14	16	18	dBm	
3) For antenna port (48~54M)	13	15	17	dBm	
2. Spectrum Mask @ target power					
1) at fc +/-11MHz	-	-	-20	dBr	
2) at fc +/-20MHz	-	-	-28	dBr	
3) at fc > +/-30MHz	-	-	-40	dBr	
3 Constellation Error(EVM)@ target power					
1) 6Mbps	-	-	-5	dB	
2) 9Mbps	-	-	-8	dB	
3) 12Mbps	-	-	-10	dB	
4) 18Mbps	-	-	-13	dB	
5) 24Mbps	-	-	-16	dB	
6) 36Mbps	-	-	-19	dB	
7) 48Mbps	-	-	-22	dB	
8) 54Mbps	-	-	-25	dB	
4 Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
5 Minimum Input Level Sensitivity					
1) 6Mbps (PER ≤10%)	-	-	-85	dBm	
2) 9Mbps (PER ≤10%)	-	-	-84	dBm	
3) 12Mbps (PER ≤10%)	-	-	-82	dBm	
4) 18Mbps (PER ≤10%)	-	-	-80	dBm	
5) 24Mbps (PER ≤10%)	-	-	-77	dBm	
6) 36Mbps (PER ≤10%)	-	-	-73	dBm	
7) 48Mbps (PER ≤10%)	-	-	-69	dBm	
8) 54Mbps (PER ≤10%)	-	-	-65	dBm	

9.3 IEEE 802.11n HT20/40 Section(2.4GHz)

Items	Contents				
Specification	IEEE802.11n HT20/40@2.4GHz				
Mode	OFDM				
Channel	HT20:CH1 to CH13 HT40:CH3 to CH11				
Data rate (MCS index)	MCS0/1/2/3/4/5/6/7				
TX Characteristics	Min.	Typ.	Max.	Unit	
1. Power Levels (Calibrated)					
1) For antenna port (MCS0~3)	16	18	20	dBm	
2) For antenna port (MCS4~5)	14	16	18	dBm	
3) For antenna port (MCS6~7)	13	15	17	dBm	
2. Spectrum Mask @target power					
1) at fc +/-22MHz	-	-	-20	dBr	
2) at fc +/-40MHz	-	-	-28	dBr	
3) at fc > +/-60MHz	-	-	-45	dBr	
3. Constellation Error(EVM)@ target power					
1) MCS0	-	-	-5	dB	
2) MCS1	-	-	-10	dB	
3) MCS2	-	-	-13	dB	
4) MCS3	-	-	-16	dB	
5) MCS4	-	-	-19	dB	
6) MCS5	-	-	-22	dB	
7) MCS6	-	-	-25	dB	
8) MCS7	-	-	-28	dB	
4. Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
5. Minimum Input Level Sensitivity			HT20	HT40	
1) MCS0 (PER ≤10%)	-	-	-82	-79	dBm
2) MCS1 (PER ≤10%)	-	-	-79	-76	dBm
3) MCS2 (PER ≤10%)	-	-	-77	-74	dBm
4) MCS3 (PER ≤10%)	-	-	-74	-71	dBm
5) MCS4 (PER ≤10%)	-	-	-70	-67	dBm
6) MCS5 (PER ≤10%)	-	-	-66	-63	dBm
7) MCS6 (PER ≤10%)	-	-	-65	-62	dBm
8) MCS7 (PER ≤10%)	-	-	-64	-61	dBm
6. Maximum Input Level (PER ≤10%)	-20	-	-	-	dBm

9.4 IEEE 802.11ax HE20/40 Section (2.4GHz)

Items	Contents				
Specification	IEEE802.11ax HE20/40@2.4GHz				
Mode	OFDMA				
Channel	HE20:CH1 to CH13 HE40:CH3 to CH11				
Data rate (MCS index)	MCS0/1/2/3/4/5/6/7/8/9/10/11				
TX Characteristics	Min.	Typ.	Max.	Unit	
1. Power Levels (Calibrated)					
1) For antenna port (MCS0~3)	16	18	20	dBm	
2) For antenna port (MCS4~5)	14	16	18	dBm	
3) For antenna port (MCS6~7)	13	15	17	dBm	
4) For antenna port (MCS8~9)	12	14	16	dBm	
5) For antenna port (MCS10~11)	11	13	15	dBm	
2. Spectrum Mask @target power					
1) at fc +/-11MHz/21MHz/41MHz	-	-	-20	dBr	
2) at fc +/-20MHz/40MHz/80MHz	-	-	-28	dBr	
3) at fc +/-30MHz/60MHz/120MHz	-	-	-40	dBr	
3. Constellation Error(EVM)@ target power					
1) MCS0	-	-	-5	dB	
2) MCS1	-	-	-10	dB	
3) MCS2	-	-	-13	dB	
4) MCS3	-	-	-16	dB	
5) MCS4	-	-	-19	dB	
6) MCS5	-	-	-22	dB	
7) MCS6	-	-	-25	dB	
8) MCS7	-	-	-27	dB	
9) MCS8	-	-	-30	dB	
10) MCS9	-	-	-32	dB	
11) MCS10	-	-	-34	dB	
12) MCS11	-	-	-35	dB	
4. Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
5. Minimum Input Level Sensitivity			HE20	HE40	
1) MCS0 (PER ≤10%)	-	-	-82	-79	dBm
2) MCS1 (PER ≤10%)	-	-	-79	-76	dBm
3) MCS2 (PER ≤10%)	-	-	-77	-74	dBm
4) MCS3 (PER ≤10%)	-	-	-74	-71	dBm
5) MCS4 (PER ≤10%)	-	-	-70	-67	dBm
6) MCS5 (PER ≤10%)	-	-	-66	-63	dBm
7) MCS6 (PER ≤10%)	-	-	-65	-62	dBm

8) MCS7 (PER ≤10%)	-	-	-64	-61	dBm	
9) MCS8(PER ≤10%)	-	-	-59	-56	dBm	
10) MCS9(PER ≤10%)	-	-	-57	-54	dBm	
11) MCS10(PER ≤10%)	-	-	-54	-51	dBm	
12) MCS11(PER ≤10%)	-	-	-51	-49	dBm	
6. Maximum Input Level (PER ≤10%)	-30	-	-	-	dBm	

9.5 IEEE 802.11a Section (5GHz)

Items	Contents				
Specification	IEEE802.11a				
Mode	OFDM				
Channel	CH36 to CH165				
Data rate (MCS index)	6, 9, 12, 18, 24, 36, 48, 54Mbps				
TX Characteristics	Min.	Typ.	Max.	Unit	
1. Power Levels (Calibrated)					
1) For antenna port (0~18M)	16	18	20	dBm	
2) For antenna port (24~36M)	14	16	18	dBm	
3) For antenna port (48~54M)	13	15	17	dBm	
2. Spectrum Mask @target power					
1) at fc +/-11MHz/21MHz/41MHz	-	-	-20	dBr	
2) at fc +/-20MHz/40MHz/80MHz	-	-	-28	dBr	
3) at fc +/-30MHz/60MHz/120MHz	-	-	-40	dBr	
3. Constellation Error(EVM)@ target power					
4) MCS3	-	-	-16	dB	
5) MCS4	-	-	-19	dB	
6) MCS5	-	-	-22	dB	
7) MCS6	-	-	-25	dB	
8) MCS7	-	-	-27	dB	
9) MCS8	-	-	-30	dB	
10) MCS9	-	-	-32	dB	
4. Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
5. Minimum Input Level Sensitivity					
1) 6Mbps (PER ≤10%)	-	-	-82	dBm	
2) 9Mbps (PER ≤10%)	-	-	-81	dBm	
3) 12Mbps (PER ≤10%)	-	-	-79	dBm	
4) 18Mbps (PER ≤10%)	-	-	-77	dBm	
5) 24Mbps (PER ≤10%)	-	-	-74	dBm	
6) 36Mbps (PER ≤10%)	-	-	-70	dBm	
7) 48Mbps (PER ≤10%)	-	-	-66	dBm	
8) 54Mbps (PER ≤10%)	-	-	-65	dBm	

6. Maximum Input Level (PER $\leq 10\%$)	-30	-	-	dBm	
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9.6 IEEE 802.11n HT20/40 Section(5GHz)

Items	Contents				
Specification	IEEE802.11n HT20/40@5GHz				
Mode	OFDM				
Channel	HT20:CH36 to CH165 HT40:CH38 to CH163				
Data rate (MCS index)	MCS0/1/2/3/4/5/6/7				
TX Characteristics	Min.	Typ.	Max.	Unit	
1. Power Levels (Calibrated)					
1) For antenna port (MCS0~3)	16	18	20	dBm	
2) For antenna port (MCS4~5)	14	16	18	dBm	
3) For antenna port (MCS6~7)	13	15	17	dBm	
2. Spectrum Mask @target power					
1) at fc +/-11MHz	-	-	-20	dBr	
2) at fc +/-20MHz	-	-	-28	dBr	
3) at fc > +/-30MHz	-	-	-45	dBr	
3. Constellation Error(EVM)@ target power					
1) MCS0	-	-	-5	dB	
2) MCS1	-	-	-10	dB	
3) MCS2	-	-	-13	dB	
4) MCS3	-	-	-16	dB	
5) MCS4	-	-	-19	dB	
6) MCS5	-	-	-22	dB	
7) MCS6	-	-	-25	dB	
8) MCS7	-	-	-28	dB	
4. Frequency Error	-20	-	20	ppm	
RX Characteristics	Min.	Typ.	Max.	Unit	
6. Minimum Input Level Sensitivity (each chain)			HT20	HT40	
1) MCS0 (PER $\leq 10\%$)	-	-	-82	-79	dBm
2) MCS1 (PER $\leq 10\%$)	-	-	-79	-76	dBm
3) MCS2 (PER $\leq 10\%$)	-	-	-77	-74	dBm
4) MCS3 (PER $\leq 10\%$)	-	-	-74	-71	dBm
5) MCS4 (PER $\leq 10\%$)	-	-	-70	-67	dBm
6) MCS5 (PER $\leq 10\%$)	-	-	-66	-63	dBm
7) MCS6 (PER $\leq 10\%$)	-	-	-65	-62	dBm
8) MCS7 (PER $\leq 10\%$)	-	-	-64	-61	dBm
6. Maximum Input Level (PER $\leq 10\%$)	-30	-	-	dBm	

9.7 IEEE 802.11ac VHT20/40/80 Section(5GHz)

Items	Contents					
Specification	IEEE802.11ac VHT20/40/80@5GHz					
Mode	OFDM					
Channel	VHT20:CH36 to CH165 VHT40:CH38 to CH163 VHT80:CH42 to CH157					
Data rate (MCS index)	MCS0/1/2/3/4/5/6/7/8/9					
TX Characteristics	Min.	Typ.	Max.			Unit
1. Power Levels (Calibrated)						
1) For antenna port (MCS0~3)	16	18	20			dBm
2) For antenna port (MCS4~5)	14	16	18			dBm
3) For antenna port (MCS6~7)	13	15	17			dBm
4) For antenna port (MCS8~9)	12	14	16			dBm
2. Spectrum Mask @target power						
1) at fc +/- 11MHz/21MHz/41MHz	-	-	-20			dBr
2) at fc +/- 20MHz/40MHz/80MHz	-	-	-28			dBr
3) at fc +/- 30MHz/60MHz/120MHz	-	-	-40			dBr
3. Constellation Error(EVM)@ target power						
1) MCS0	-	-	-5			dB
2) MCS1	-	-	-10			dB
3) MCS2	-	-	-13			dB
4) MCS3	-	-	-16			dB
5) MCS4	-	-	-19			dB
6) MCS5	-	-	-22			dB
7) MCS6	-	-	-25			dB
8) MCS7	-	-	-27			dB
9) MCS8	-	-	-30			dB
10) MCS9	-	-	-32			dB
4. Frequency Error	-20	-	20			ppm
RX Characteristics	Min.	Typ.	Max.			Unit
5.Minimum Input Level Sensitivity (each chain)			VHT20	VHT40	VHT80	
1) MCS0 (PER ≤10%)	-	-	-82	-79	-76	dBm
2) MCS1 (PER ≤10%)	-	-	-79	-76	-73	dBm
3) MCS2 (PER ≤10%)	-	-	-77	-74	-71	dBm
4) MCS3 (PER ≤10%)	-	-	-74	-71	-68	dBm
5) MCS4 (PER ≤10%)	-	-	-70	-67	-64	dBm
6) MCS5 (PER ≤10%)	-	-	-66	-63	-60	dBm
7) MCS6 (PER ≤10%)	-	-	-65	-62	-59	dBm
8) MCS7 (PER ≤10%)	-	-	-64	-61	-58	dBm
9) MCS8(PER ≤10%)	-	-	-59	-56	-53	dBm

10) MCS9(PER $\leq 10\%$)	-	-	-57	-54	-51	dBm	
6. Maximum Input Level (PER $\leq 10\%$)	-30	-	-		dBm		

9.8 IEEE 802.11ax HE20/40/80 Section(5GHz)

Items	Contents					
Specification	IEEE802.11ax HE20/40/80@5GHz					
Mode	OFDMA					
Channel	HE20:CH36 to CH165 HE40:CH38 to CH163 HE80:CH42 to CH157					
Data rate (MCS index)	MCS0/1/2/3/4/5/6/7/8/9/10/11					
TX Characteristics	Min.	Typ.	Max.		Unit	
1. Power Levels (Calibrated)						
1) For antenna port (MCS0~3)	16	18	20		dBm	
2) For antenna port (MCS4~5)	14	16	18		dBm	
3) For antenna port (MCS6~7)	13	14	17		dBm	
4) For antenna port (MCS8~9)	12	13	16		dBm	
5) For antenna port (MCS10~11)	10	12	14		dBm	
2. Spectrum Mask @target power						
1) at fc +/-11MHz/21MHz/41MHz	-	-	-20		dBr	
2) at fc +/-20MHz/40MHz/80MHz	-	-	-28		dBr	
3) at fc +/-30MHz/60MHz/120MHz	-	-	-40		dBr	
3. Constellation Error(EVM)@ target power						
1) MCS0	-	-	-5		dB	
2) MCS1	-	-	-10		dB	
3) MCS2	-	-	-13		dB	
4) MCS3	-	-	-16		dB	
5) MCS4	-	-	-19		dB	
6) MCS5	-	-	-22		dB	
7) MCS6	-	-	-25		dB	
8) MCS7	-	-	-27		dB	
9) MCS8	-	-	-30		dB	
10) MCS9	-	-	-32		dB	
11) MCS10	-	-	-34		dB	
12) MCS11	-	-	-35		dB	
4. Frequency Error	-20	-	20		ppm	
RX Characteristics	Min.	Typ.	Max.		Unit	
5. Minimum Input Level Sensitivity			HE20	HE40	HE80	
1) MCS0 (PER $\leq 10\%$)	-	-	-82	-79	-76	dBm
2) MCS1 (PER $\leq 10\%$)	-	-	-79	-76	-73	dBm
3) MCS2 (PER $\leq 10\%$)	-	-	-77	-74	-71	dBm

4) MCS3 (PER $\leq 10\%$)	-	-	-74	-71	-68	dBm	
5) MCS4 (PER $\leq 10\%$)	-	-	-70	-67	-64	dBm	
6) MCS5 (PER $\leq 10\%$)	-	-	-66	-63	-60	dBm	
7) MCS6 (PER $\leq 10\%$)	-	-	-65	-62	-59	dBm	
8) MCS7 (PER $\leq 10\%$)	-	-	-64	-61	-58	dBm	
9) MCS8 (PER $\leq 10\%$)	-	-	-59	-56	-53	dBm	
10) MCS9 (PER $\leq 10\%$)	-	-	-57	-54	-51	dBm	
11) MCS10 (PER $\leq 10\%$)	-	-	-54	-51	-49	dBm	
12) MCS11 (PER $\leq 10\%$)	-	-	-51	-49	-46	dBm	
6. Maximum Input Level (PER $\leq 10\%$)	-30	-	-	-	-	dBm	

9.9 Bluetooth Section

Items	Contents					
Specification	BT2.1+EDR/4.2/5.2/5.3					
Mode	FHSS,GFSK,DPSK,DQPSK					
Number of Channel	79 Channels					
Frequency Band	2.402 GHz ~2.480GHz					
	Min.	Typ.	Max.	Unit	Remark	
1. Output Power	-	7	-	dBm		
2. Gain step	-	1.0	-	dB		
3. Receiver sensitivity (BER $\leq 0.1\%$)						
1).Basic data rate receiver specifications (BER $\leq 0.1\%$)	-	-92	-80			
2).Enhanced data rate receiver specifications (BER $\leq 0.1\%$)	-	-93	-80			
3).Bluetooth LE receiver specifications (PER $\leq 30.8\%$)	-	-95	-80			
4. Maximum usable signal (BER $\leq 0.1\%$)	-	-5	-			
5. C/I co-channel (BER $<0.1\%$)	-	4	11	dB		
6. C/I 1MHz (BER $<0.1\%$)	-	-14	0	dB		
7. C/I 2MHz (BER $<0.1\%$)	-	-42	-30	dB		
8. C/I ≥ 3 MHz (BER $<0.1\%$)	-	-49	-40	dB		
9. C/I Image channel (BER $<0.1\%$)	-	-25	-9	dB		
10. C/I Image 1MHz (BER $<0.1\%$)	-	-50	-20	dB		
11. Inter-modulation	-	-13	-	dB		
12. Out-of-band blocking						
1). 30MHz to 2000MHz	-10	-	-	dBm		
2). 2000MHz to 2399MHz	-27	-	-	dBm		
3). 2498MHz to 3000MHz	-27	-	-	dBm		
4). 3000MHz to 12.75GHz	-10	-	-	dBm		
13. Modulation characteristics						

1). Δf_{1avg}				KHz	
2). Δf_{2max} (For at least 99.9% of all Δf_{2max})	-	-	-	KHz	
3). $\Delta f_{1avg} / \Delta f_{2avg}$	-	-	-	KHz	
14. ICFT	-	-	-	KHz	
15. Carrier frequency drift					
1). One slot packet (DH1)	-25	± 15	+25	KHz	
2). Two slot packet (DH3)	-40	± 15	+40	KHz	
3). Five slot packet (DH5)	-40	± 15	+40	KHz	
4). Max drift rate	-	6	20	KHz/50us	
16. TX output spectrum(20dB bandwidth)	-	922	1000	KHz	
17. In-Band spurious emission					
1). ± 2 MHz offset	-	-45	-20	dBm	
2). ± 3 MHz offset	-	-48	-40	dBm	
3). $> \pm 3$ MHz offset	-	-48	-40	dBm	

10. Reference Design (参考设计)

10.1 DC Electrical Characteristics (直流电气特性)

Symbol	Description	conditions	Min.	Typ.	Max.	Unit
VDD33	Power supplies	-	3.0	3.3	3.6	V
VDDIO	I/O input power supplies	-	3.0	3.3	3.6	V
			1.7	1.8	1.9	
IvDD33	Power supply current	-	-	-	800	mA
IvDDIO	I/O supply current	-	-	-	50	mA
V _{IH}	High-level input voltage	VDDIO=3.3V VDDIO=1.8V	VDDIO*0.625 VDDIO*0.65	-	VDDIO+0.3	V
V _{IL}	Low-level input voltage	VDDIO=3.3V VDDIO=1.8V	-0.3	-	VDDIO*0.25 VDDIO*0.35	V
V _{OH}	High-level output voltage	VDDIO=3.3V VDDIO=1.8V	VDDIO-0.4 VDDIO-0.2	-	VDDIO+0.3	V
V _{OL}	Low-level output voltage	VDDIO=3.3V VDDIO=1.8V	-0.3	-	0.4 0.2	V
R _{PU}	Internal pull-up resistor	VDDIO=3.3V VDDIO=1.8V	40 10	75 50	190 100	k Ω
R _{PD}	Internal pull-down resistor	VDDIO=3.3V VDDIO=1.8V	40 10	75 50	190 100	k Ω

10.2 Power On Sequence (上电时序)

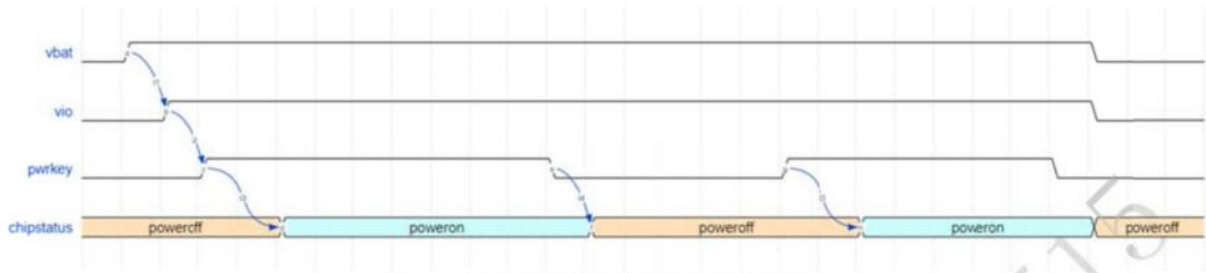
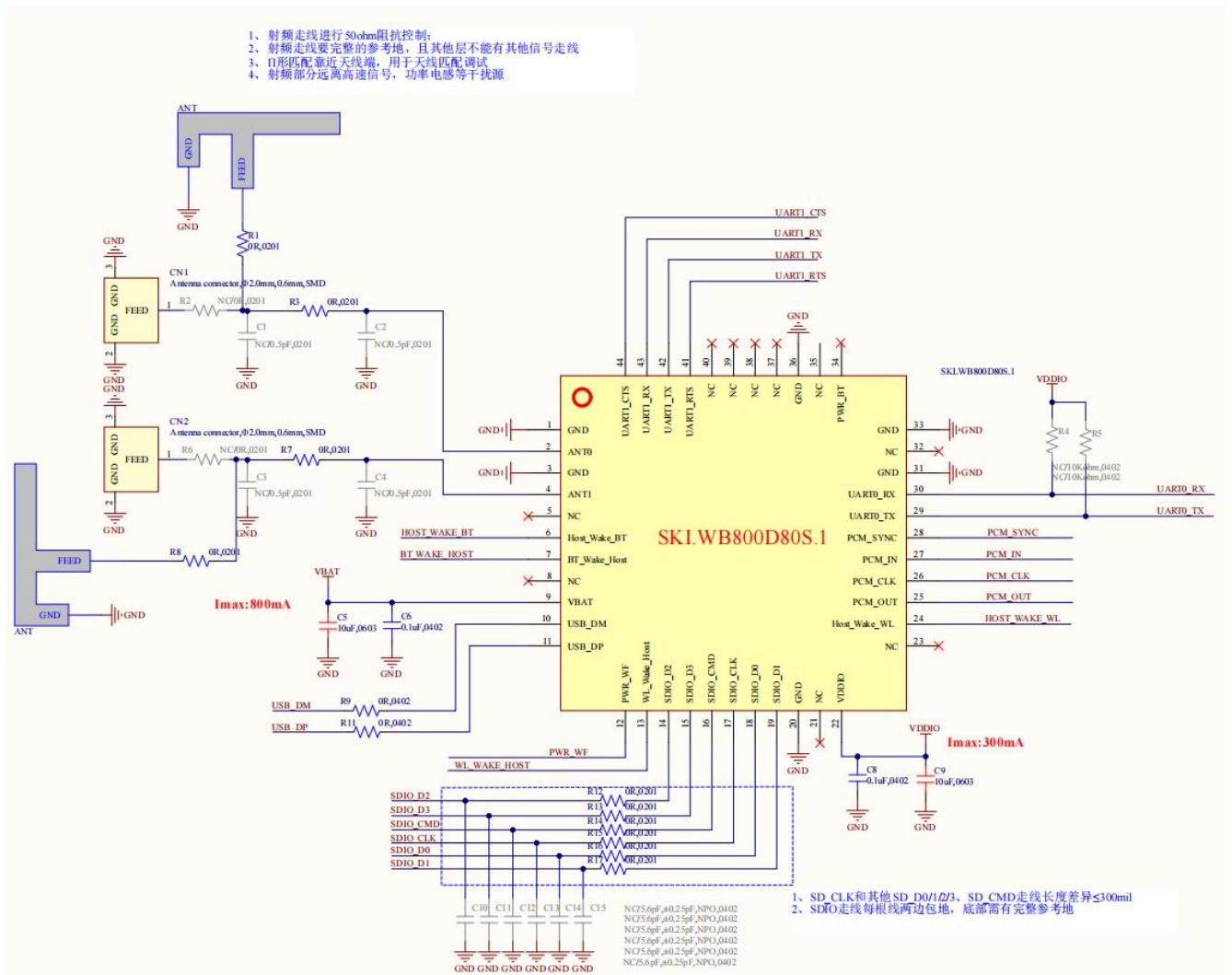


Figure 5-1 AIC8800D80 上电时序

- t1: VIO's power on time $\geq$ VBAT's
- t2: power key's high time $\geq$ VIO's +200us
- t3: chip all power on ready time $\geq$ power key high time + 8ms
- t4: pwrkey pull low to chip all power off time $\geq$ 6ms

10.3 Reference schematic (参考原理图)



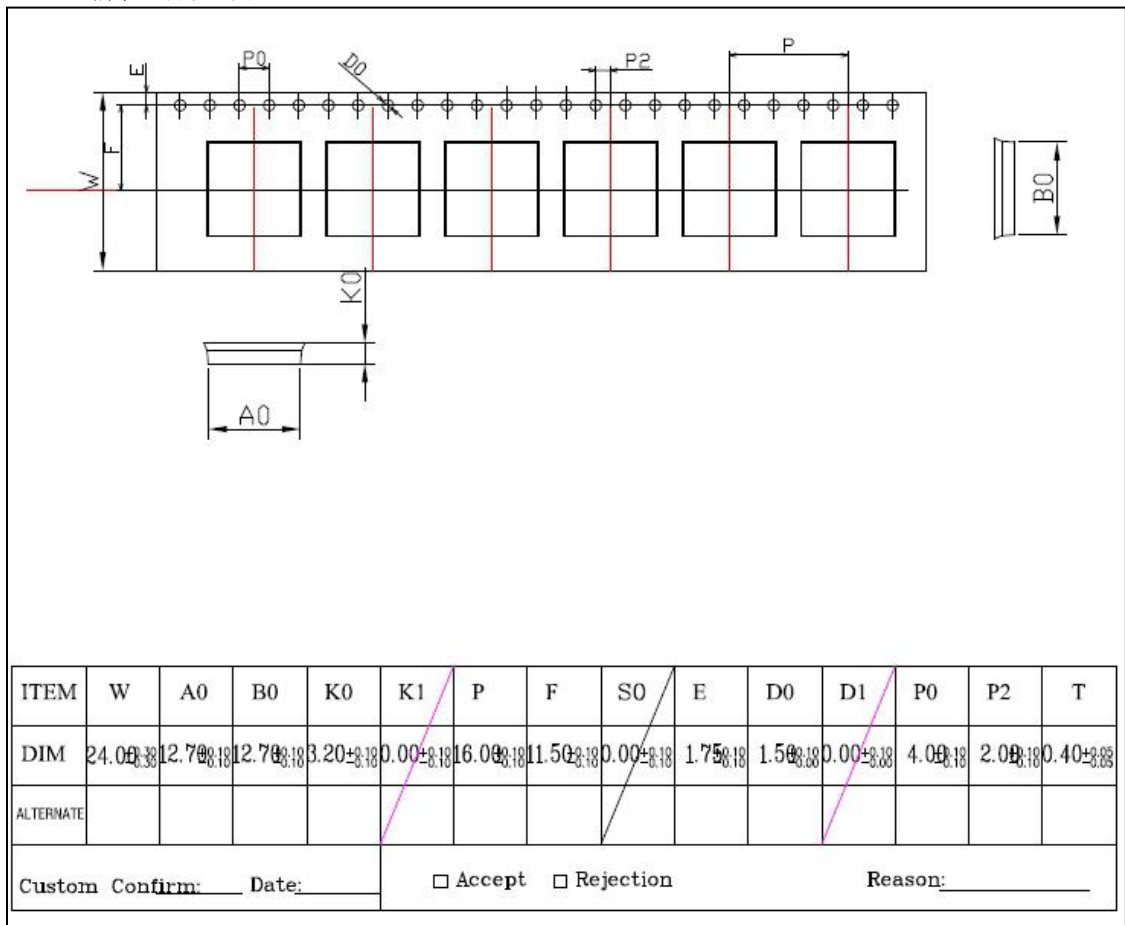
11. Mechanical, Environmental and Reliability Tests

Test Items		Test Conditions	Qty	Criteria Condition
11-1	Drop test	The packed samples within 100Kg can be tested Drop height: Face Side: 800/600/450mm Edge line: 600/450/350mm Drop time: 1 each Face and edge.	1xBox	After drop test, the outer box and inner box will not be broken by appearance visual inspection.
11-2	Vibration test	X-Y-Z direction, first Frequency changing from 10Hz to 30Hz to 10Hz, amplitude 0.75mm, 5 times vibrations, then frequency Changing from 30Hz to 55 Hz to 30 Hz, amplitude 0.15mm, 5 time vibration.	3	After test, the Appearance, Power EVM and Frequency error shall be satisfied with the specification.
11-3	Impact test	Impact acceleration: 50m/sec ² ; Impact duration: 16ms; Impact times: 1000.	3	After test, the Appearance, Power EVM and Frequency error shall be satisfied with the specification.
11-4	Soldering ability test	Soldering temperature: 235±5℃ Soldering duration: 2±0.5S	3	1. After soldering, the soldered area must be covered by a smooth bright solder layer, some deficiencies such as a small amount of the pinhole, not wetting are allowed, but the deficiencies can not be in the same place; 2. At least 90% of soldered area shall be covered continuously by the soldering material.
11-5	Humidity test	Leave samples in 40±3℃, 93% RH @ 96 hours	3	Leave samples in standard test condition for 2 hours then test, the Appearance, Power, EVM and Frequency error functional parameter shall be satisfied with the test specification.

11-6	High temperature load life test	Thermostat cabinet temperature: 55±5℃ Applied voltage: 110% rated voltage Working duration: 200 hour (Supply Voltage Cycle 23h power on, 1h power off)	60	After test, leave samples in standard condition for 1 hour and test, Power, EVM and Frequency error shall be satisfied with the test specification.
11-7	High temperature load test	Temperature: 55±5℃ Samples work for 16 hours	3	After test, the Appearance, Power, EVM and Frequency error shall be Satisfied with the test specification.
11-8	Low temperature storage test	Leave the samples in -25±3℃@24 hours	3	Leave samples in standard test condition for 2 hours then test, the Appearance, Power, EVM and Frequency error shall be satisfied with the test specification.
11-9	Low temperature load test	Leave samples in -15±3℃@ 2 hours, samples' function shall be normal, the let samples work for 1 hour	3	After test, leave the samples in standard condition and tested the Appearance, Power, EVM and Frequency error shall be satisfied with the test specification.
11-10	Temperature circle test	One cycle duration -10±3℃@3H 40±3℃ @3H Total cycle: 10x	3	After test, leave the samples in standard condition and tested Power EVM and Frequency error shall be qualified and all the characters shall be satisfied with the test specification.
11-11	Continuous TP test	Twice cycle duration -10±3℃@4H +60±3℃@4H, +25@2H@2H	3	During test, There will not been appeared signal disconnection or interruption between DUT and AP.
11-12	ESD	Discharge voltage: 1kV C: 150pF Discharge resistance: 330Ω Positive10 times 1 time for each second	3	The products can recoverable smoothly after ESD test.

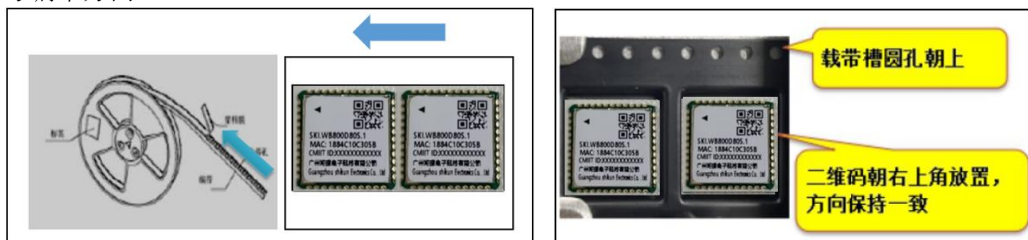
12. Package (包装)

(1) 编带包装示意图



(2) 编带方向

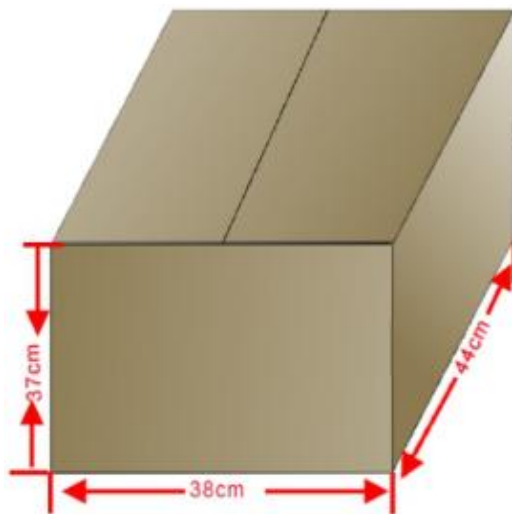
带屏蔽罩料号编带方向:



不带屏蔽罩料号编带方向:



(3) 外箱图纸示意图



(4) 包装要求信息

每盘包装1500pcs，每箱装8盘，每箱数量=1500*8=12000pcs

13. 存储、生产

13.1 存储要求

本产品的湿敏特性为4级（MSL4），出厂时以真空密封袋包装。产品搬运、存储、加工过程必须遵循 IPC/JEDEC J-STD-033。在环境温度低于 40 摄氏度，空气湿度小于90%的情况下，真空包装下产品可存放12个月。在产品存放有效期内，如发现真空包装有漏气、湿敏卡变色达到烘烤标准、开封暴露时间超过72H，需要烘烤后使用。

13.2 生产参数

炉温最高不能超过250℃，推荐是240℃。
推荐SMT焊炉温曲线如下图所示。

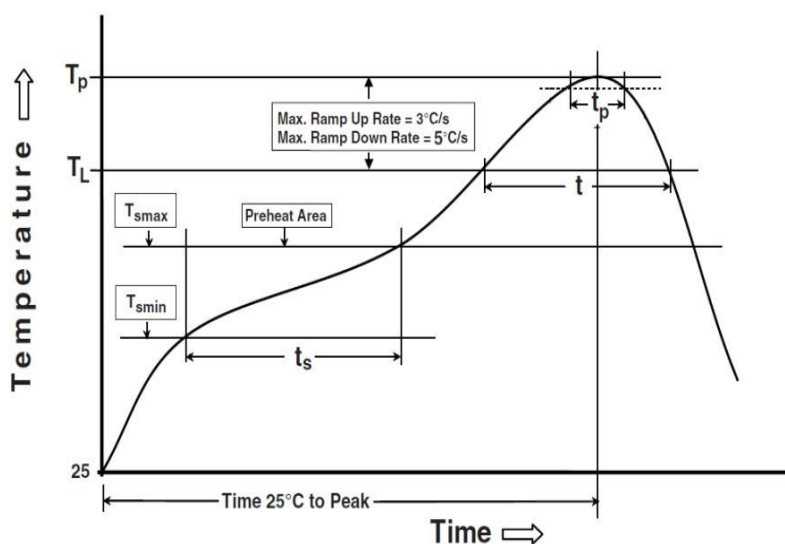


图 13-1 炉温曲线推荐图

炉温参数	最小值	典型值	最大值	单位
预热区最低温度 T_{smin}	150			°C
预热区最高温度 T_{smax}			200	°C
预热上升时间 t_s	80		120	s
回流焊区升温速率(TL 到 T_p)			3	°C/s
回流焊区低温 T_L		220		°C
回流焊区峰值温度 T_p	235	240	250	°C
回流焊峰值温度时间 t_p (T_p 波动 5°C 范围)			30	s
回流焊区冷却降温速率 (T_p 到 T_L)	-5	-3	-1	°C/s
回流时间 t	40		60	s

图 13-2 炉温曲线参数